

Netflix Movie Analysis Report

1. Project Overview

This project analyzes movies in Netflix using movies dataset that consists 9000+ movies according to popularity, genre and voting. The goal is to unfold the insights of the movies in Netflix according to its genre and popularity to give proper review of the movies that goes trending in Netflix.

2. Dataset summary

The dataset consists of:

Rows: 9827

Columns: 9

The features in the dataset:

- Movie description (title, overview, language)
- Movie details (genre, popularity, release date)
- Movie Real-Time-Voting (vote count, vote average)
- There are 0 missing values in the dataset

3. Exploratory Data Analysis (EDA):

The EDA in the Netflix Movie Dataset is done using python and its libraries. Pandas is used for Data cleaning and matplotlib and seaborn is used for Data visualization.

1. Imported Dataset into jupyterlab using pandas and made a copy of the original dataset for further data cleaning

```
# Loading the dataset and checking the data
df = pd.read_csv('Netflix_Movie_Dataset/netflix_movie_analysis.csv', lineterminator= "\n")
df_copy = df.copy()
✓ 0.0s
```

2. Loaded the first 5 data in the dataset to check if it has successfully been implemented.

```
# Loading the first 5 data of the dataset
df_copy.head()
```

✓ 0.0s

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language	Genre
0	2021-12-15	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en	Action, Adventure, Science Fiction
1	2022-03-01	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en	Crime, Mystery, Thriller
2	2022-02-25	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en	Thriller
3	2021-11-24	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en	Animation, Comedy, Family, Fantasy
4	2021-12-22	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en	Action, Adventure, Thriller, War

	Poster_Url
	https://image.tmdb.org/t/p/original/1g0dhYtq4i...
	https://image.tmdb.org/t/p/original/74xTEgt7R3...
	https://image.tmdb.org/t/p/original/vDHsLnOWKI...
	https://image.tmdb.org/t/p/original/4j0PNHkMr5...
	https://image.tmdb.org/t/p/original/aq4Pwv5Xeu...

3. Checked the null values in the dataset and info

```
# Checking the null values in the dataset
df_copy.isnull().sum()

✓ 0.0s
```

Release_Date	0
Title	0
Overview	0
Popularity	0
Vote_Count	0
Vote_Average	0
Original_Language	0
Genre	0
Poster_Url	0
dtype: int64	

```
# viewing the info of the dataset
df_copy.info()

✓ 0.0s
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Release_Date          9827 non-null   str
1   Title                 9827 non-null   str
2   Overview              9827 non-null   str
3   Popularity            9827 non-null   float64
4   Vote_Count            9827 non-null   int64
5   Vote_Average          9827 non-null   float64
6   Original_Language     9827 non-null   str
7   Genre                 9827 non-null   str
8   Poster_Url            9827 non-null   str
dtypes: float64(2), int64(1), str(6)
memory usage: 691.1 KB
```

4. Checked the genres column to explore the data present in genres

```
# exploring genres column
df_copy['Genre'].head()

✓ 0.0s
```

0	Action, Adventure, Science Fiction
1	Crime, Mystery, Thriller
2	Thriller
3	Animation, Comedy, Family, Fantasy
4	Action, Adventure, Thriller, War

Name: Genre, dtype: str

5. Checked the duplicates in the dataset

```
# Checking the duplicated rows
df_copy.duplicated().sum()

✓ 0.0s
```

np.int64(0)

6. Checked the summary statistics of the dataset

```
# Checking summary statistics
```

```
df_copy.describe()
```

✓ 0.0s

	Popularity	Vote_Count	Vote_Average
count	9827.000000	9827.000000	9827.000000
mean	40.326088	1392.805536	6.439534
std	108.873998	2611.206907	1.129759
min	13.354000	0.000000	0.000000
25%	16.128500	146.000000	5.900000
50%	21.199000	444.000000	6.500000
75%	35.191500	1376.000000	7.100000
max	5083.954000	31077.000000	10.000000

7. Converted release date from object to datetime and again changed the datatype of datetime

```
# Converting the obj of release_date to date_time
```

```
df_copy["Release_Date"] = pd.to_datetime(df_copy["Release_Date"])
```

```
print(df_copy["Release_Date"].dtypes)
```

✓ 0.0s

datetime64[us]

```
# Changing the datatype of datetime
```

```
df_copy["Release_Date"] = df_copy["Release_Date"].dt.year
```

```
df_copy["Release_Date"].dtypes
```

✓ 0.0s

dtype('int32')

8. Checked the updated info of the dataset

```
df_copy.info()
```

✓ 0.0s

```
<class 'pandas.DataFrame'>  
RangeIndex: 9827 entries, 0 to 9826  
Data columns (total 9 columns):  
#   Column                Non-Null Count  Dtype  
---  -  
0   Release_Date          9827 non-null   int32  
1   Title                  9827 non-null   str  
2   Overview               9827 non-null   str  
3   Popularity             9827 non-null   float64  
4   Vote_Count             9827 non-null   int64  
5   Vote_Average           9827 non-null   float64  
6   Original_Language      9827 non-null   str  
7   Genre                  9827 non-null   str  
8   Poster_Url             9827 non-null   str  
dtypes: float64(2), int32(1), int64(1), str(5)  
memory usage: 652.7 KB
```

9. Dropped overview, original language and poster URL since these data won't affect our movie analysis

```
# dropping overview, original_language, poster_url
cols = ["Overview", "Original_Language", "Poster_Url"]

df_copy.drop(columns= cols, inplace=True)
df_copy.columns

✓ 0.0s

Index(['Release_Date', 'Title', 'Popularity', 'Vote_Count', 'Vote_Average',
      'Genre'],
      dtype='str')
```

```
df_copy.head()

✓ 0.0s
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	8.1	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	6.3	Thriller
3	2021	Encanto	2402.201	5076	7.7	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	7.0	Action, Adventure, Thriller, War

10. Replaced the data in vote average into 4 different categories according to its summary

```
# categorizing vote_average column
def categorize_column(df_copy, col, label):
    edges = [df_copy[col].describe()["min"],
             df_copy[col].describe()["25%"],
             df_copy[col].describe()["50%"],
             df_copy[col].describe()["75%"],
             df_copy[col].describe()["max"]]
    df_copy[col] = pd.cut(df_copy[col], edges, labels= label, duplicates='drop')
    return df_copy

labels = ["not_popular", "below_avg", "average", "popular"]
categorize_column(df_copy, "Vote_Average", label= labels)

# Confirming changes
df_copy["Vote_Average"].unique()

✓ 0.0s
```

```
['popular', 'below_avg', 'average', 'not_popular', NaN]
Categories (4, str): ['not_popular' < 'below_avg' < 'average' < 'popular']
```

```
df_copy.head()
```

```
✓ 0.0s
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	popular	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	below_avg	Thriller
3	2021	Encanto	2402.201	5076	popular	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	average	Action, Adventure, Thriller, War

11. Checked the null values in the updated vote average and dropped the null values if it exists

```
# exploring columns

df_copy["Vote_Average"].value_counts()
✓ 0.0s

Vote_Average
not_popular    2467
popular        2450
average        2412
below_avg      2398
Name: count, dtype: int64

df_copy.isnull().sum()
✓ 0.0s

Release_Date    0
Title           0
Popularity      0
Vote_Count      0
Vote_Average    100
Genre           0
dtype: int64

#Dropping null values
df_copy.dropna(inplace=True)
df_copy.isna().sum()
✓ 0.0s

Release_Date    0
Title           0
Popularity      0
Vote_Count      0
Vote_Average    0
Genre           0
dtype: int64
```

12. Spitted and exploded the genre column and casted it as type category

```
# splitting the genre
df_copy["Genre"] = df_copy["Genre"].str.split(", ")

# explode the lists
df_copy = df_copy.explode("Genre").reset_index(drop=True)
df_copy.head()
```

✓ 0.0s

Release_Date		Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction
3	2022	The Batman	3827.658	1151	popular	Crime
4	2022	The Batman	3827.658	1151	popular	Mystery

```
# casting column into category
df_copy["Genre"] = df_copy["Genre"].astype("category")

# confirming changes
df_copy["Genre"].dtypes
```

✓ 0.0s

```
CategoricalDtype(categories=['Action', 'Adventure', 'Animation', 'Comedy', 'Crime',
                             'Documentary', 'Drama', 'Family', 'Fantasy', 'History',
                             'Horror', 'Music', 'Mystery', 'Romance', 'Science Fiction',
                             'TV Movie', 'Thriller', 'War', 'Western'],
                  ordered=False, categories_dtype=str)
```

13. Checked the final updated info and checked the distinct data in the dataset

```
df_copy.info()
```

✓ 0.0s

```
<class 'pandas.DataFrame'>  
RangeIndex: 25552 entries, 0 to 25551  
Data columns (total 6 columns):  
#   Column          Non-Null Count  Dtype  
---  -  
0   Release_Date    25552 non-null  int32  
1   Title           25552 non-null  str  
2   Popularity      25552 non-null  float64  
3   Vote_Count      25552 non-null  int64  
4   Vote_Average    25552 non-null  category  
5   Genre           25552 non-null  category  
dtypes: category(2), float64(1), int32(1), int64(1), str(1)  
memory usage: 749.1 KB
```

```
df_copy.nunique()
```

✓ 0.0s

```
Release_Date    100  
Title           9415  
Popularity      8088  
Vote_Count      3265  
Vote_Average     4  
Genre           19  
dtype: int64
```

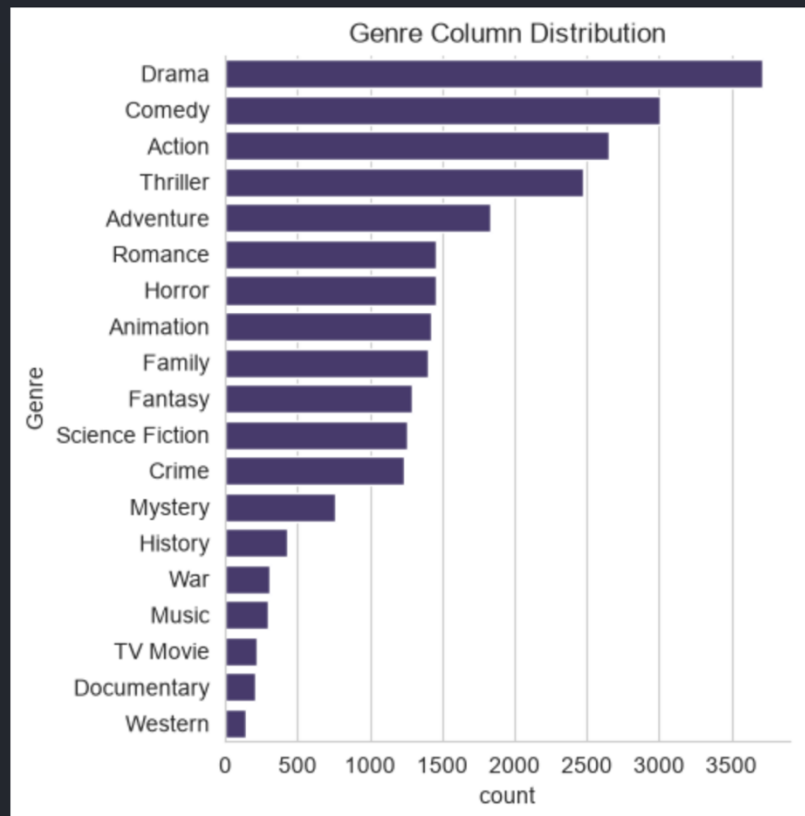
4. Data Visualization:

The data visualization is done using matplotlib and seaborn library in jupyterlab after data cleaning using pandas.

1. Category chart of Genre column distribution

```
# vizualizing genre column
sns.catplot(y= "Genre", data= df_copy, kind='count',
            order=df_copy["Genre"].value_counts().index,
            color = "#433273")
plt.title(label="Genre Column Distribution")
plt.show()
```

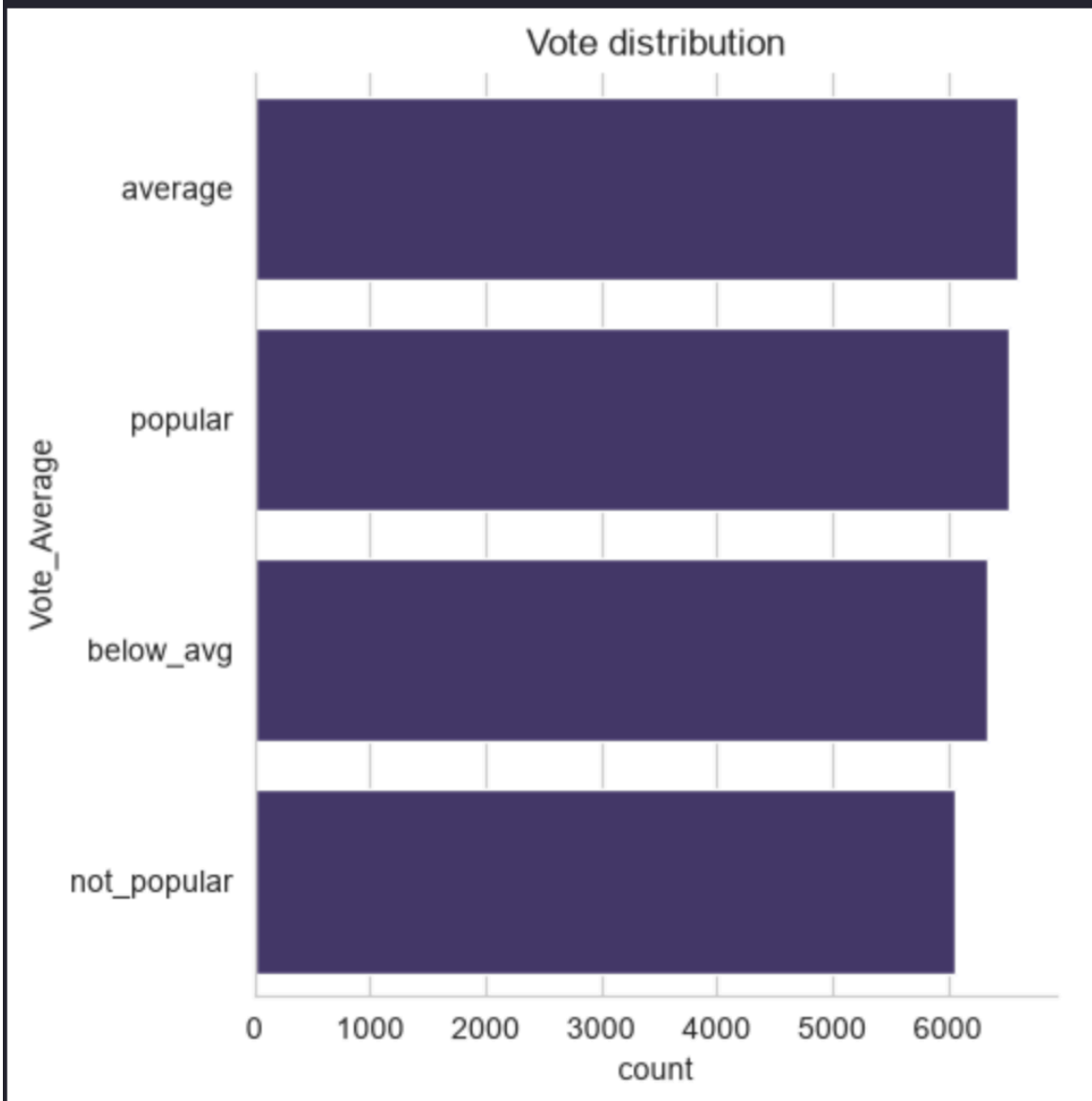
✓ 0.1s



2. Vote distribution chart using vote average

```
#vizualizing vote_average coloumn
sns.catplot(data=df_copy, y="Vote_Average", kind="count",
            order= df_copy["Vote_Average"].value_counts().index,
            color="#433273")
plt.title("Vote distribution")
plt.show()
```

✓ 0.0s



3. Maximum popularity in the dataset

```
# Checking the max value in the popularity column
# What movie got the highest popularity and what is its genre.
df_copy[df_copy["Popularity"] ==df_copy["Popularity"].max()]
✓ 0.0s
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction

4. Minimum popularity in the dataset

```
# Checking the max value in the popularity column
# What movie got the highest popularity and what is its genre.
df_copy[df_copy["Popularity"] ==df_copy["Popularity"].max()]
✓ 0.0s
```

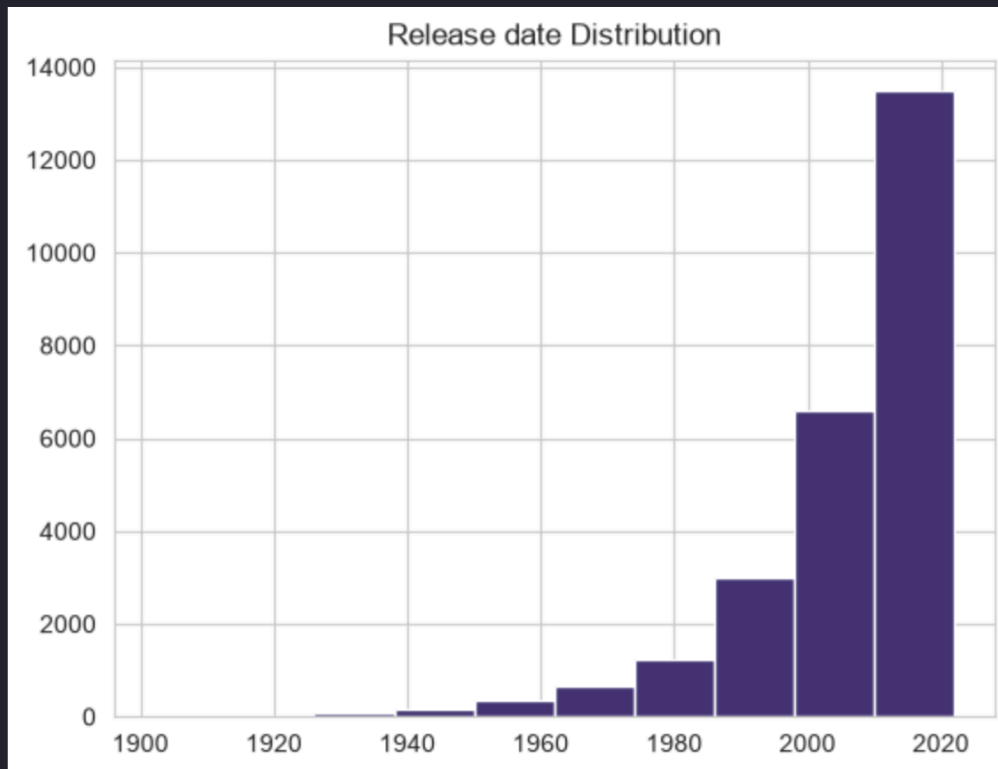
	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction

5. Release date distribution chart

```
#vizualizing release_date column
```

```
plt.hist(x= df_copy["Release_Date"], color="#433273")  
plt.title("Release date Distribution")  
plt.show()
```

✓ 0.0s



5. Conclusion:

Q1: What is the most frequent genre in the dataset?

Drama genre is the most frequent genre in our dataset and has appeared more than 14% of the times among 19 other genres.

Q2: What genres has highest votes ?

we have 25.5% of our dataset with popular vote (6520 rows). Drama again gets the highest popularity among fans by being having more than 18.5% of movies popularities.

Q3: What movie got the highest popularity ? what's its genre ?

Spider-Man: No Way Home has the highest popularity rate in our dataset and it has genres of Action , Adventure and Science Fiction.

Q4: What movie got the lowest popularity ? what's its genre ?

The United States, thread' has the highest lowest rate in our dataset and it has genres of music , drama , 'war', 'sci-fi' and history`.

Q5: Which year has the most filmed movies?

year 2020 has the highest filming rate in our dataset.

